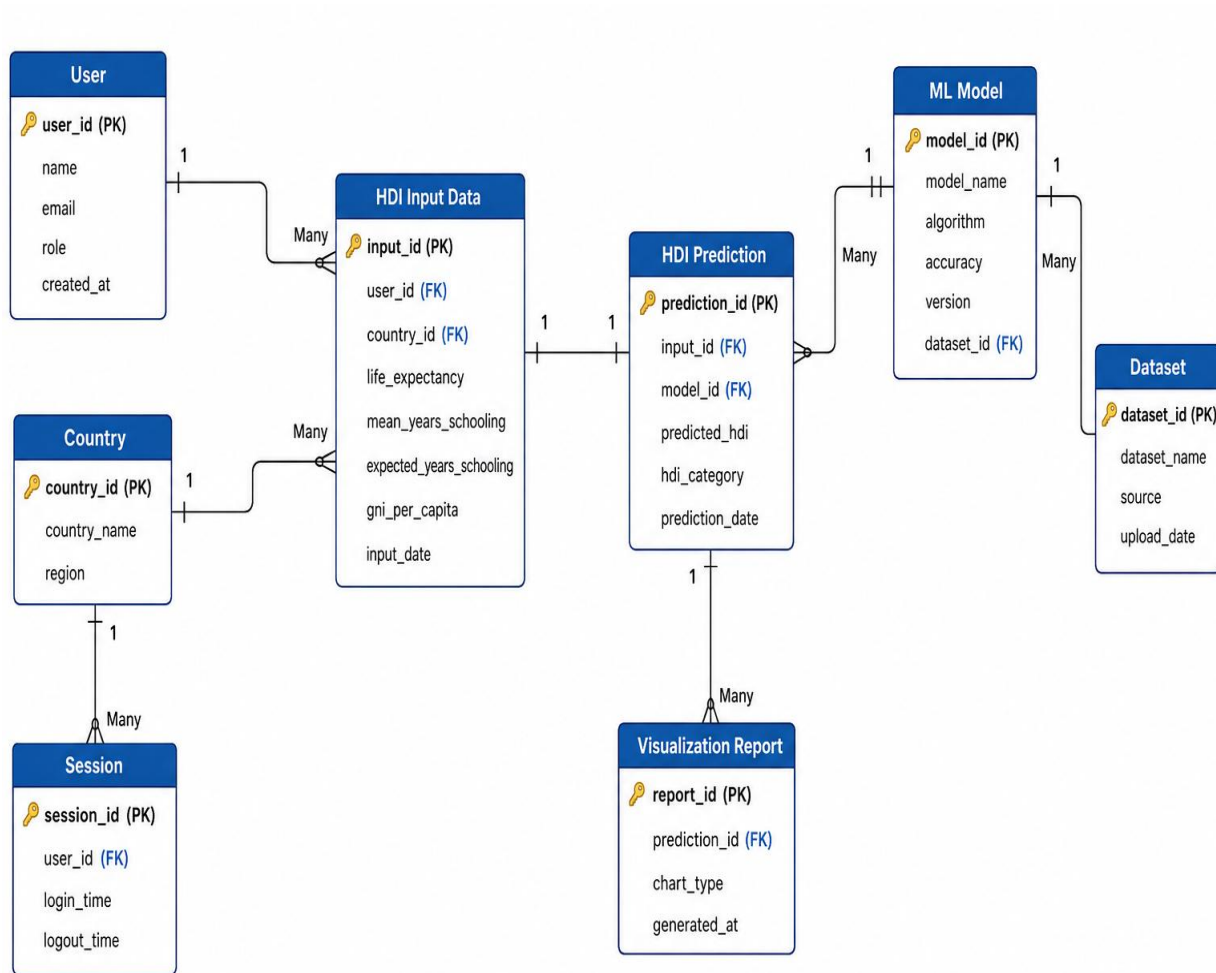


Entity-Relationship Diagram

Date	26 June 2026
Project Name	A Comprehensive Measure of Well-Being

Entity-Relationship Diagram



Overview

The Entity-Relationship (ER) diagram represents the core entities involved in the A Comprehensive Measure of Well-Being system. It defines how data related to users, countries, HDI inputs, predictions, machine learning models, datasets, and visual reports interact within the system.

This structure supports storing and analyzing Human Development Index (HDI)–related data and generating predictions and visual insights using machine learning models.

Entities Involved

The system consists of eight primary entities:

- User
- Country
- HDI Input Data
- ML Model
- HDI Prediction
- Dataset
- Visualization Report
- Session

Primary Keys

Each entity is uniquely identified using a primary key:

- User → user_id
- Country → country_id
- HDI Input Data → input_id
- ML Model → model_id
- HDI Prediction → prediction_id
- Dataset → dataset_id
- Visualization Report → report_id
- Session → session_id

Relationships

- User → HDI Input Data: One user can submit multiple HDI input records (1:M)
- Country → HDI Input Data: One country can have multiple HDI input records (1:M)
- HDI Input Data → HDI Prediction: Each input generates one prediction (1:1)
- ML Model → HDI Prediction: One model can generate multiple predictions (1:M)
- HDI Prediction → Visualization Report: One prediction can generate multiple reports (1:M)
- Dataset → ML Model: One dataset can train multiple models (1:M)
- User → Session: One user can have multiple sessions (1:M)

User Attributes

The User entity contains:

- user_id (Primary Key)
- name

- email
- role
- created_at

Summary

This ER structure ensures organized storage and processing of HDI-related data, enabling accurate prediction generation, model training, and visualization of well-being metrics across countries.